

Smart Helmet for Accident Prevention and Detection

This project is an **IoT-based Smart Helmet** designed to improve the safety of two-wheeler riders. The main purpose is to **prevent accidents, reduce drunk driving, ensure helmet usage, detect accidents automatically, and immediately inform emergency contacts with the accident location.**

What Problem Does This Project Solve?

Road accidents are increasing every year due to several reasons:

- Riders do not wear helmets.
- Riders drive after consuming alcohol.
- Accidents are reported late.
- Ambulances and emergency services cannot reach the accident location quickly.
- Family members do not know where the accident occurred.
- Delayed medical assistance leads to unnecessary deaths.

This project solves these problems by combining smart sensors, wireless communication, GPS, and cloud technology into a smart helmet system.

Main Idea of the Project

The system is divided into **two parts**:

1. Helmet Unit (Transmitter)

Mounted inside the helmet.

2. Bike Unit (Receiver)

Mounted on the motorcycle.

Both units communicate wirelessly using an RF/Zigbee module. The system continuously checks whether the rider is wearing the helmet, whether alcohol is detected, and whether an accident has occurred. If an accident happens, it sends the rider's GPS location to emergency contacts.

Features of the Project

1. Helmet Detection

Problem

Many riders ignore wearing helmets.

Solution

A touch/force sensor is placed inside the helmet.

When the rider wears the helmet:

- Sensor detects the head.
- Signal is sent to the bike.

If the helmet is not worn:

- Bike ignition remains OFF.

Benefit

- Prevents riding without a helmet.
 - Improves rider safety.
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2. Alcohol Detection

Problem

Drunk driving causes many fatal accidents.

Solution

An MQ-2 gas sensor is placed near the rider's mouth.

It detects alcohol in the rider's breath.

If alcohol is detected:

- Bike will not start.

Benefit

- Prevents drunk driving.
 - Reduces accident risk.
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3. Bike Ignition Control

The bike starts only if:

✓ Helmet is worn

AND

✓ Alcohol is not detected

Otherwise:

✗ Ignition remains OFF.

This feature ensures riders follow basic safety rules before riding.

4. Accident Detection

Problem

After an accident, nobody knows immediately.

Solution

Gyroscope and vibration sensors continuously monitor the bike.

If an accident occurs:

- Sensors detect abnormal vibration and movement.
- Accident signal is generated automatically.

Benefit

- No manual action required.
 - Faster emergency response.
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5. GPS Tracking

GPS module continuously tracks the bike location.

If an accident occurs:

- Exact latitude and longitude are obtained.

This helps emergency services quickly locate the rider.

6. GSM Communication

After accident detection:

GSM module sends:

- SMS
- Phone Call
- GPS Location

to registered mobile numbers.

This allows family members or emergency responders to receive immediate information.

7. IoT Cloud Monitoring

The system uploads collected data to the cloud.

Users can:

- Monitor system status

- Track information remotely
 - Access accident information using IoT
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8. Wireless Communication

Helmet communicates with the bike using:

- RF Module
- Zigbee (NRF24L01)

No wired connection is required.

Components Used

Component	Purpose
Arduino UNO	Main Controller
ADXL335 Accelerometer/Gyroscope	Accident Detection
Touch Sensor	Helmet Detection
MQ-2 Gas Sensor	Alcohol Detection
Vibration Sensor	Accident Detection
GPS Module	Live Location

GSM Module

SMS & Calling

NRF24L01 (Zigbee/RF)

Helmet-Bike Communication

Relay

Bike Ignition Control

DC Motor (Prototype)

Simulated Bike Engine

LCD Display

Displays System Status

These components are described throughout the methodology section.

System Workflow

Wear Helmet



Touch Sensor Checks Helmet



MQ2 Checks Alcohol



Helmet Sends Data to Bike



Arduino Processes Data



Helmet Worn?



YES

Alcohol Detected?



NO

Bike Ignition ON

Advantages

- Prevents riding without a helmet.
- Prevents drunk driving.
- Automatic accident detection.
- Live GPS tracking.
- Automatic emergency SMS and phone calls.
- Low-cost IoT solution.
- Improves rider safety.
- Reduces emergency response time.

These capabilities are the intended outcomes described by the paper.

Limitations

According to the paper:

- Sensors may be damaged during a severe accident.
 - If hardware fails, false or no readings may occur.
 - The system depends on hardware sensors for detection.
 - Additional improvements are needed for more reliable accident prevention.
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Future Scope

The authors suggest several future improvements:

- Camera-based vision system for monitoring rider behavior.
 - Better accident prevention techniques that do not rely entirely on physical sensors.
 - Vehicle-to-Vehicle (V2V) wireless communication.
 - Improved traffic monitoring by authorities.
 - Enhanced communication between nearby vehicles.
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Final Summary

This project is an **IoT-based safety system for motorcycles** that combines a smart helmet with a bike-mounted controller. Before the ride, it verifies that the rider is wearing a helmet and has not consumed alcohol. During the ride, it monitors for accidents using motion and vibration sensors. If an accident occurs, it automatically obtains the rider's GPS location and sends alerts via GSM to emergency contacts while updating the information to the cloud. Overall, the system aims to reduce accident-related fatalities by promoting safer riding and enabling faster emergency response.